
NFL PLAYOFF VS NON- PLAYOFF TEAMS

Uzair Ahmed

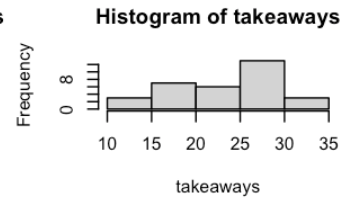
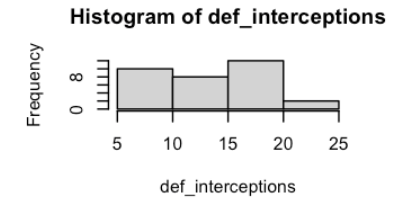
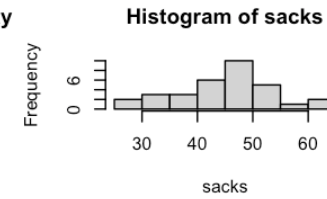
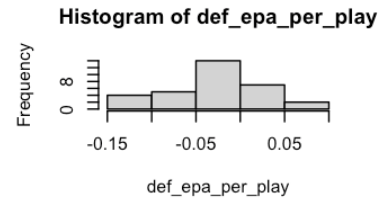
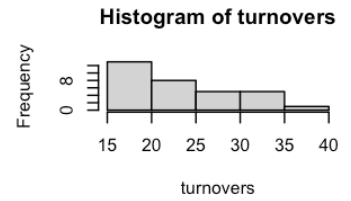
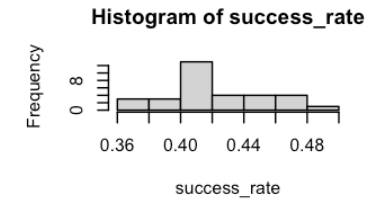
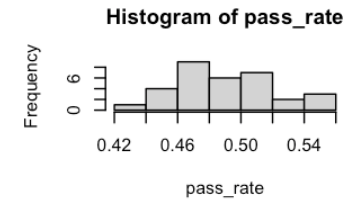
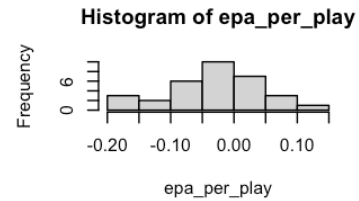


INTRODUCTION

- The goal of my project is to compare how NFL playoff teams differ from non-playoff teams across various performance and efficiency stats
 - The data was taken from the nflfastR package and data was taken from the 2023 season using the pbp and team stats functions
 - The 8 stats I decided to look at were EPA per Play, Success Rate, Pass Rate, Turnovers, Defensive EPA per Play, Sacks, Interceptions and Takeaways
 - EPA: advanced and efficiency metric, expected points added, expected points describes how many points, on average, a team is expected to score on a possession given a particular field position.
 - Success Rate: rate at which a play is successful, play deemed successful if they gain at least 50% of yards on 1st down, 70% on 2nd down and 100% on 3rd down
-

UNIVARIATE ANALYSIS

Variables	Skewness	Kurtosis
EPA per Play	-0.22275421	-0.4790754
Pass Rate	0.29349009	-0.7568451
Success Rate	0.38712294	-0.6352754
Turnovers	0.54400636	-0.7995997
Def EPA per Play	0.03894328	-0.4674180
Sacks	-0.12662929	-0.3123911
Interceptions	0.28911987	-0.8239809
Takeaways	-0.30938801	-0.7448635



NORMALITY

- Multivariate normality was assessed using the Henze-Zirkler test
- The test returned a p-value of 0.216 meaning we fail to reject the null hypothesis and conclude the data is multivariately normal
- Univariate Anderson–Darling tests showed a mild deviation in 2 variables: success rate and turnovers but MVN was still satisfied overall

```
$multivariate_normality
      Test Statistic p.value    Method    MVN
1 Henze-Zirkler    0.949    0.216 asymptotic ✓ Normal
```

ONE WAY MANOVA & UNIVARIATE ANOVAS

One Way MANOVA			
	Wilks	F	P-value
playoff	0.39466	19.788	0.00241

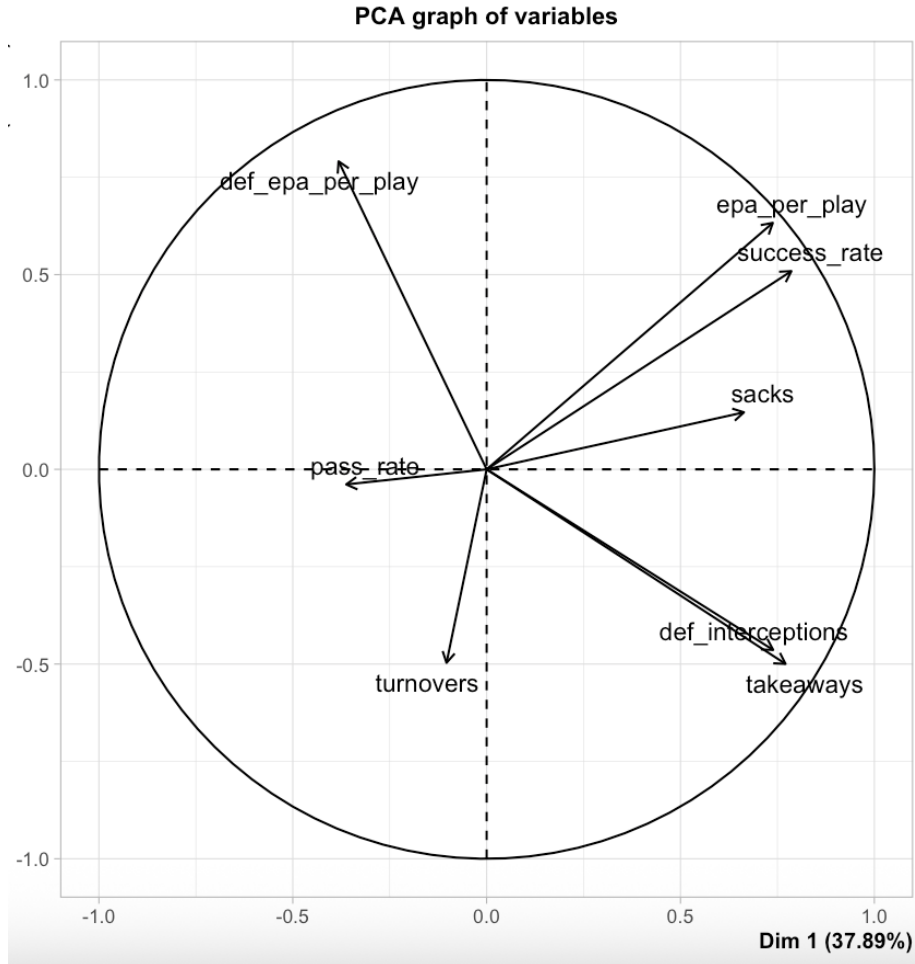
Univariate ANOVAS		
Variables	P-value	Significant
EPA per Play	0.00011	Yes
Pass Rate	0.1874	No
Success Rate	0.0002	Yes
Turnovers	0.9424	No
Def EPA per Play	0.2907	No
Sacks	1.665e-05	Yes
Interceptions	0.1242	No
Takeaways	0.03822	Yes

PCA ANALYSIS

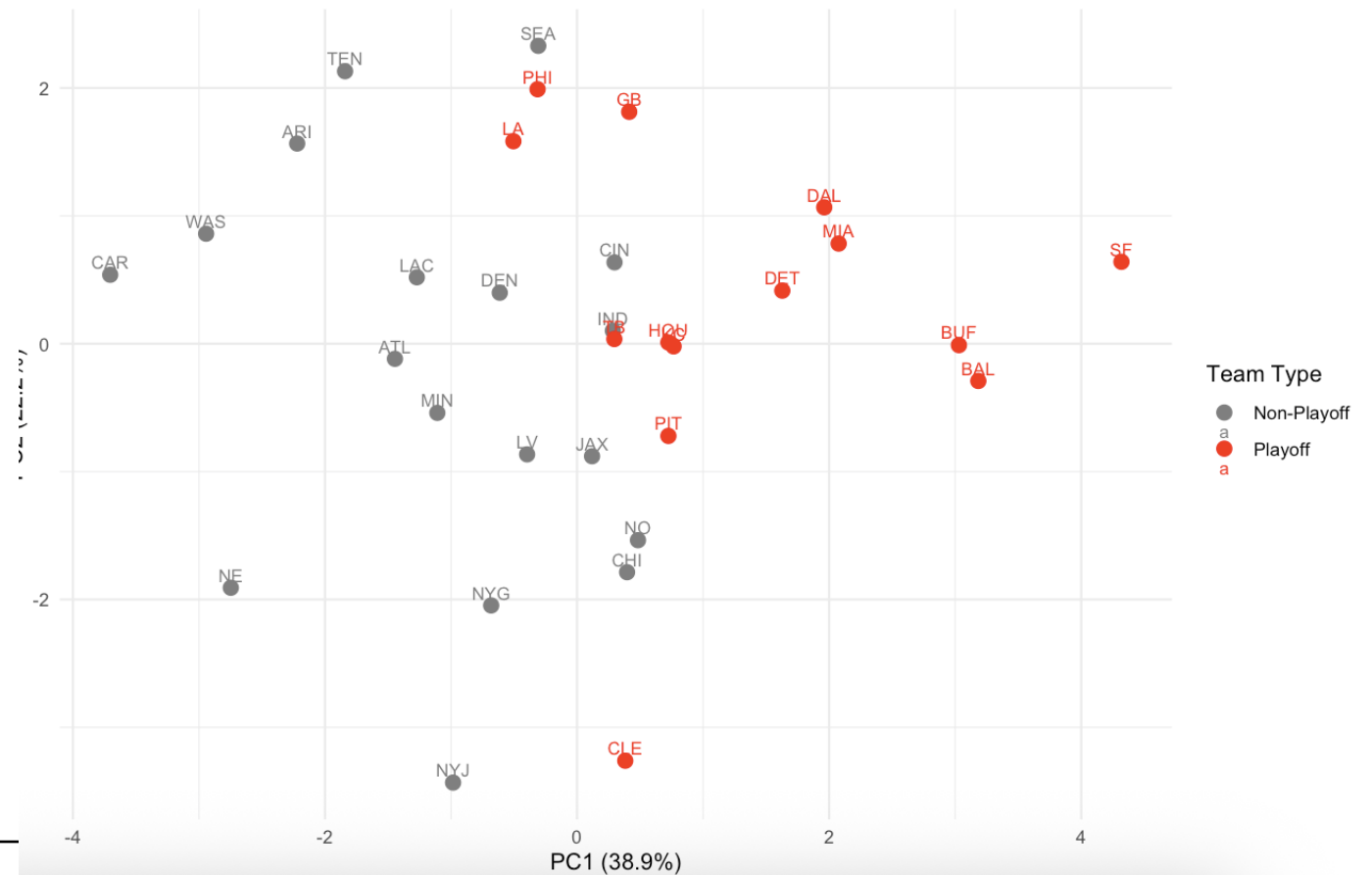
PCA		
	% of Variance	Cumulative %
PC 1	38.89068	38.89068
PC 2	25.24670	63.13739
PC 3	13.24377	79.38116
PC 4	9.21192	88.59309
PC 5	6.55749	95.15058
PC 6	3.33979	98.49037
PC 7	1.07157	99.56195
PC 8	0.43805	100

PCA Loadings		
Variables	PC1	PC2
EPA per Play	0.7381290	0.63296874
Pass Rate	-0.3621368	-0.03875974
Success Rate	0.7859988	0.50901455
Turnovers	-0.1034988	-0.49684061
Def EPA per Play	-0.3818763	0.79087902
Sacks	0.6636285	0.14644586
Interceptions	0.7390078	-0.46372198
Takeaways	0.7709774	-0.49966397

PCA ANALYSIS PLOTS



Playoff vs Non Playoff on PCA Dimensions



LOGISTIC REGRESSION

	Estimate	Std Error	Z	P Value	Odds Ratio
Intercept	-0.5196	0.5942	-0.875	0.3818	--
PC1	2.8214	1.2806	2.203	0.0276	16.7998022
PC2	0.5850	0.4193	1.395	0.1630	1.7949516

Null deviance: 43.860 on 31 degrees of freedom

Residual deviance: 19.331 on 29 degrees of freedom AIC: 25.331

		0	1
Confusion Matrix:	0	16	3
	1	2	11

Correctly Predicted: $27/32 = 84.375\%$
Misclassification Rate: $5/32 = 15.625\%$

CONCLUSIONS & LIMITATIONS

- Overall, playoff and non-playoff teams differed multivariately across the 8 performance metrics examined
 - PCA showed that most of this separation occurs along the first principal component, which represented overall team efficiency and defensive disruption
 - Given the increase in passing volume was surprised pass rate was not as significant
 - Some limitations include:
 - Sample size is small ($n = 32$ teams for one season) which increases the chance of noise and reduces statistical power
 - Results are based on a single season and relationships may vary across years.
 - The logistic model may overfit slightly due to limited sample size.
 - Additional seasons or cross-validation would provide stronger evidence.
-